EX. NO: 10

INTERNETWORKING WITH ROUTERS; WIRELESS ROUTER, DHCP SERVER AND INTERNET CLOUD

10(a) INTERNETWORKING WITH A ROUTER

Aim: To design a network using one router connecting two separate LANs.

Device	IP Address
Router Gig0/0	10.1.1.1
Router Gig0/1	10.2.2.1
PC0	10.1.1.10
PC1	10.2.2.10

1. Place a router and two PCs in Packet Tracer.
2. Connect PC0 to Gig0/0 and PC1 to Gig0/1.
3. Configure the router interfaces with the IP addresses given above.
4. Set IP address and gateway on each PC.
5. Ping from PC0 to PC1 to check connectivity.

Result: Connectivity between PC0 and PC1 was verified successfully.

10(b) WIRELESS ROUTER, DHCP SERVER AND INTERNET CLOUD

Aim: To configure a wireless network with automatic IP addressing using DHCP and connect to an outside server through Internet Cloud.

1. Place a wireless router, DHCP server, laptops, a web server and an Internet Cloud.
2. Enable DHCP service on the server with the pool 192.168.5.100 to 192.168.5.150.
3. Set each laptop's IP configuration to DHCP.
4. Connect laptops to the wireless network using the SSID and password.
5. Ping the outside web server to check connectivity through the cloud.

Result: The laptops received IP addresses automatically from the DHCP server, and the outside web server was reached successfully through the Internet Cloud.